Entailment as unsatisfiability of the negated extension

The Open Logic Project contributors

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-first-order-logic-syntax-and-semantics/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof uses only satisfaction definitions - no citations - so stubs are zero; the 'to the contrary' supposition structure is verbatim.

Editorial note. The label prop:entails-unsat names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own prop environment in the unit source.

Allowed standard prerequisites (editorial).

- Definitions of entailment, joint satisfaction, unsatisfiability. Opening of the same section: Gamma entails A iff every structure satisfying all of Gamma satisfies A; a set is unsatisfiable with no model. Both directions are pure unfoldings of these clauses; negated-satisfaction is read as failure of satisfaction.

Proposition 0.1. $\Gamma \models A$ *iff* $\Gamma \cup \{\neg A\}$ *is unsatisfiable.*

Proof. For the forward direction, suppose $\Gamma \models A$ and suppose to the contrary that there is structure $\mathfrak{M}$ so that $\mathfrak{M} \vDash \Gamma \cup \{\neg A\}$. Since $\mathfrak{M} \vDash \Gamma$ and $\Gamma \models A$, $\mathfrak{M} \vDash A$. Also, since $\mathfrak{M} \vDash \Gamma \cup \{\neg A\}$, $\mathfrak{M} \vDash \neg A$, so we have both $\mathfrak{M} \vDash A$ and $\mathfrak{M} \nvDash A$, a contradiction. Hence, there can be no such structure $\mathfrak{M}$, so $\Gamma \cup \{\neg A\}$ is unsatisfiable.

For the reverse direction, suppose $\Gamma \cup \{\neg A\}$ is unsatisfiable. So for every structure $\mathfrak{M}$, either $\mathfrak{M} \nvDash \Gamma$ or $\mathfrak{M} \vDash A$. Hence, for every structure $\mathfrak{M}$ with $\mathfrak{M} \vDash \Gamma$, $\mathfrak{M} \vDash A$, so $\Gamma \models A$. □